

Waseem Raza, PhD

Machine Learning Engineer | AI/ML, GenAI & Wireless Systems

Applied AI | Generative & Agentic AI | Data Science | Wireless Networks
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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering and 7+ years of combined industry, research, and engineering experience developing end-to-end AI/ML systems. Expertise includes Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Generative AI, retrieval-augmented generation (RAG), domain adaptation, time-series modeling, data engineering, Azure Databricks, Apache Spark, MLOps, and wireless-network intelligence. Experienced in translating ambiguous technical requirements into validated machine-learning workflows, production-oriented data pipelines, and measurable decision-support systems.

Technical Skills

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, classification, regression, anomaly detection, transfer learning, domain adaptation, time-series modeling, cross-validation, and model evaluation

Generative & Agentic AI: RAG, embeddings, vector retrieval, prompt engineering, structured extraction, tool-using agents, LangChain, LangGraph, agent orchestration, and grounded response generation

Data Engineering & Analytics: Python, SQL, Pandas, NumPy, ETL/ELT, feature engineering, data validation, Azure Databricks, Apache Spark, PySpark, and cloud analytics workflows

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, model deployment, and reproducible workflows

Wireless Systems: LTE/4G, 5G-NR, NR-NTN, mobility, RAN optimization, propagation modeling, network measurements, and service assurance

Wireless AI & Reliability: Network intelligence, outage management, transfer learning, adversarial learning, generative augmentation, self-healing, and resilient automation

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

02/2025 - Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer

Impac Exploration Services

09/2024 - 12/2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Advanced-Degrees Intern <i>AT&T Labs</i>	06/2024 - 08/2024 <i>Austin, TX</i>
- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%. - Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation. - Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.	
Advanced Wireless Technology Development Intern <i>Futurewei Technologies</i>	09/2022 - 01/2023 <i>Schaumburg, IL</i>
- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets. - Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions. - Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.	
Advanced-Degrees Intern <i>AT&T Labs</i>	06/2022 - 08/2022 <i>Bedminster, NJ</i>
- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%. - Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership. - Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.	
Graduate Research Assistant <i>AI4Networks, University of Oklahoma</i>	09/2019 - 07/2025 <i>Tulsa, OK</i>
- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed. - Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling. - Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation. - Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%. - Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.	

Education

PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	<i>09/2019 - 07/2025</i>
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	<i>08/2014 - 08/2016</i>
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	<i>11/2010 - 06/2014</i>

Projects

Agentic AI and RAG Applications <i>Portfolio Project</i>	<i>2026 In progress</i>
Practical experiments with retrieval, tool-using agents, orchestration, memory, evaluation, and multi-agent workflows. <i>Tools & methods:</i> LangGraph, CrewAI, AutoGen, RAG, Vector Databases	
AI-Ready Lakehouse Pipeline <i>Portfolio Project</i>	<i>2024</i>
An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads. <i>Tools & methods:</i> Azure, Databricks, Spark, Data Engineering	
SyntheticNET: 3GPP-Compliant Network Simulation <i>Research Project</i>	<i>2020-2024</i>
Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks. <i>Tools & methods:</i> Python, 3GPP, Simulation, Wireless Networks	
Wearable Intelligence for Health Monitoring <i>Research Project</i>	<i>2024-2026</i>
Deep-learning methods for continuous recognition of medication and dietary activities from wrist-worn inertial sensors. <i>Tools & methods:</i> Wearable AI, Time Series, Deep Learning, Smartwatch Sensing	
System Design Case-Study Portfolio <i>Guided Project</i>	<i>2026 In progress</i>
Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs. <i>Tools & methods:</i> System Design, APIs, Distributed Systems, Scalability	

Certifications

1. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

Focus: Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

Tools & Skills: Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development
2. IBM RAG and Agentic AI Professional Certificate — [IBM / Coursera](#) | Completed (8/8) courses | 02/2026 | [Certificate](#)

Original eight-course credential.

Focus: Completed eight-course professional certificate covering retrieval-augmented generation, vector databases, multimodal applications, and agentic AI workflows.

Tools & Skills: Retrieval-Augmented Generation, Agentic AI, LangChain, LangGraph, Vector Databases, ChromaDB, FAISS, LlamaIndex, Multimodal AI, Tool Calling, Prompt Engineering, CrewAI, AutoGen / AG2, BeeAI, Gradio, Python, Flask
3. Machine Learning — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization
4. Google Cloud Data Analytics Professional Certificate — Google Cloud / Coursera | Completed

Focus: Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.

Tools & Skills: SQL, BigQuery, Data Analytics, Visualization
5. AI for Telecommunications Specialization — [AI CERTs / Coursera](#) | Completed (3/3) courses | 06/2026 | [Certificate](#)

Focus: Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.

Tools & Skills: AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow
6. 4G Network Fundamentals — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | 08/2026 | [Certificate](#)

Focus: Foundational study of fourth-generation cellular networks.

Tools & Skills: 4G Networks, Cellular Networks, Telecommunications

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches	2026
<i>HealthINF Accepted</i>	
Wearable AI for continuous, real-world medication-intake monitoring.	
On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks	2025
<i>Journal article</i>	
Data-driven optimization and proactive reliability for future cellular-network operation.	
An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin	2024
<i>IEEE PIMRC</i>	
Robust, data-driven propagation modeling for wireless-network digital twins.	